

Statistical Baseline Model

TASK – 1

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APMS 2014 Mental Health (MH) Survey

1. Data Overview :

- Data Source : NHS Digital
- Dataset Size : The dataset consists of 702 records and 15 variables.
- Types of variables : The dataset includes categorical variables such as Region, Disorder, and Sex, along with numerical variables such as prevalence rates, confidence intervals, and sample size.

2. Data Description :

- Used APMS 2014 Chapters Files to extract tables details.
- Types of Disorder : CMD, PTSD, Psychotic Disorder, Personality Disorder, ADHD, Bipolar, Alcohol Use, Drug Dependence, Suicidal Behaviour.
- Each chapters consists of multiple table (2.1, 2.2...) from which different values were exrtacted.

3. Summary Statistics : We analyze numerical columns to observe :

- Mean
- Min
- Max
- average prevalence
- highest prevalence
- variability

- skewness indications

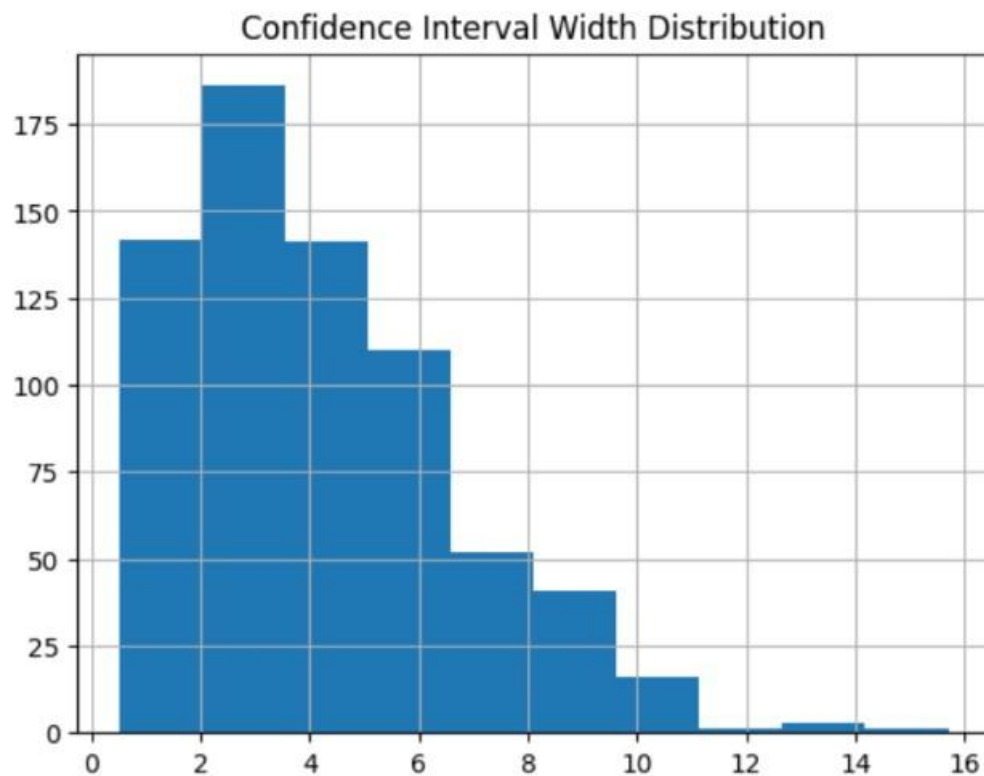
	Prevalence_AgeStd_%	Sample_size	Standard_error%	CI_Width_pp
count	693.000000	702.000000	693.000000	693.000000
mean	10.286363	566.994302	1.069223	4.191276
std	16.289624	291.486558	0.633947	2.485036
min	0.069957	132.000000	0.129100	0.506200
25%	1.572806	343.000000	0.571900	2.241900
50%	4.450085	513.000000	0.970400	3.804000
75%	12.312600	740.000000	1.392400	5.457900
max	90.736010	2429.000000	4.006700	15.706200

4. Confidence Interval (CI) :

Our data is present in the form of percentage so, we use Wilson Score Method to calculate our confidence interval.

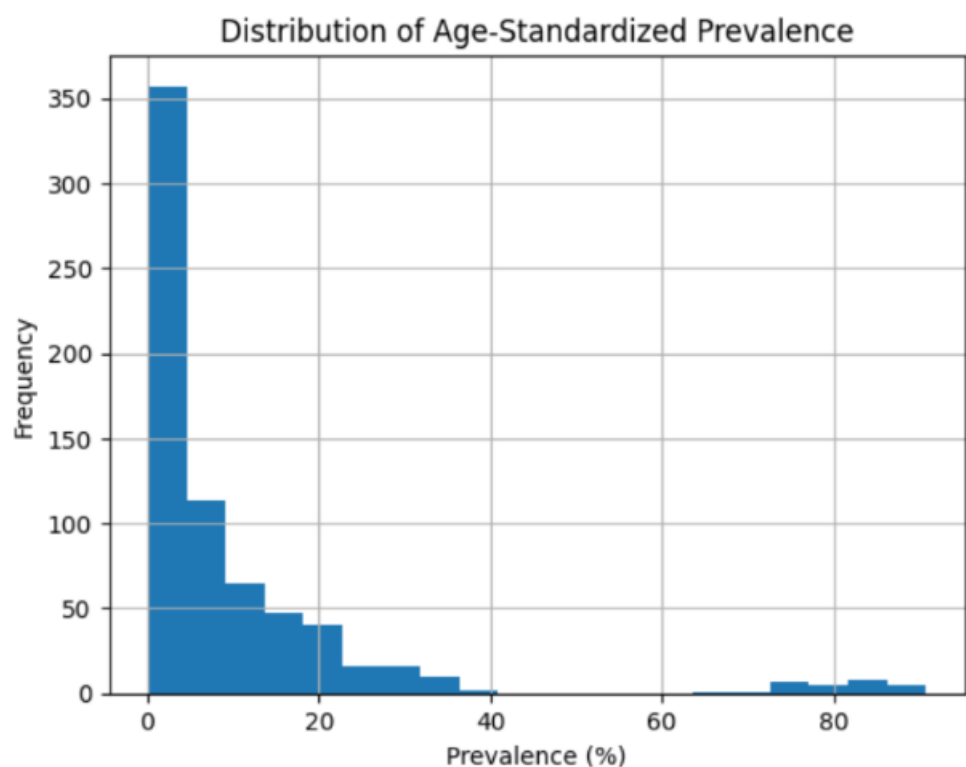
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We can see in the below given graph that the narrow CI width shows more precise estimate, less uncertainty and often caused by large sample sizes whereas wide CI width shows more uncertainty and less precision



5. Distribution of Prevalence Rates :

The histogram of prevalence rates was used to examine the distribution of mental health prevalence across observations. The graph illustrates the frequency distribution of prevalence estimates and helps identify skewness, clustering patterns, and potential outliers. Most prevalence values were concentrated within a moderate range, while a smaller number of observations showed comparatively higher prevalence levels.



6 . Region-wise Statistical Tables:

Generated separate tables for:

1. CMD
2. PTSD
3. Psychotic Disorder
4. Personality Disorder
5. Bipolar Disorder
6. Drug Dependence

Each Table include:

1. Geographic region
2. Prevalence %
3. Sample size (n)
4. CI lower & upper
5. CI width

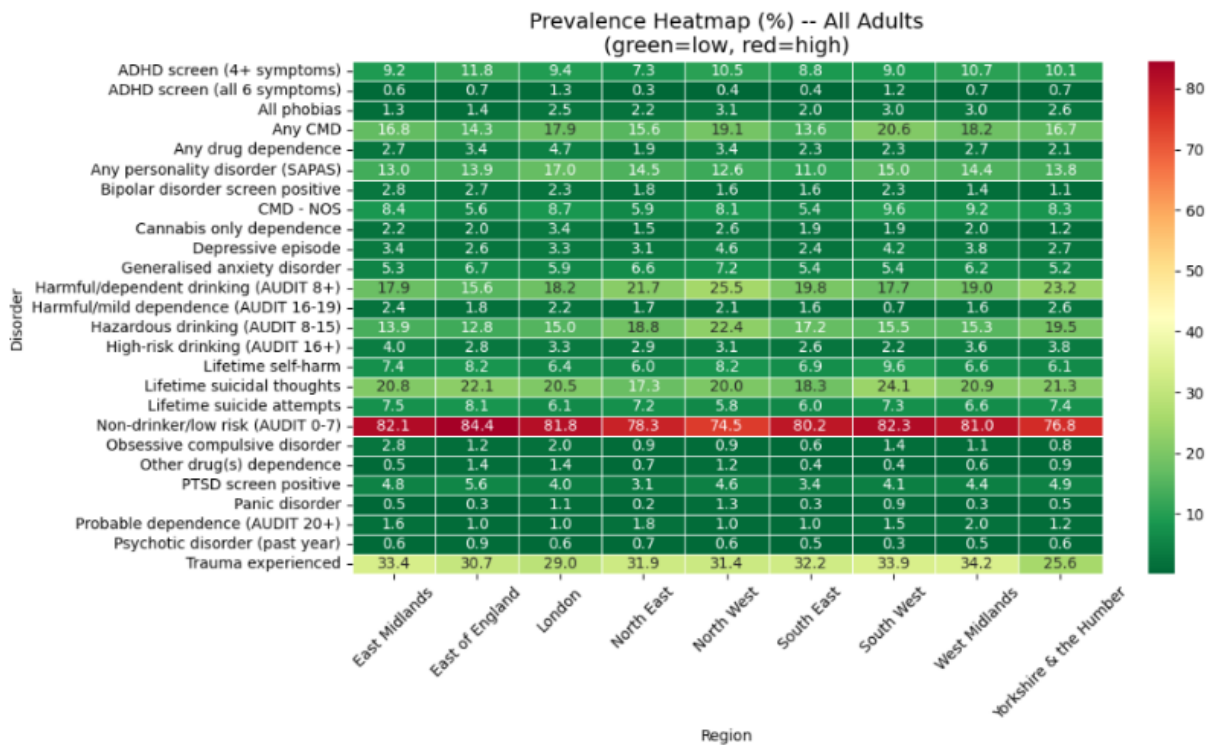
6. Standard error(%)

COMMON MENTAL DISORDERS					
Region	Prevalence%	n	95% CI Lower	95% CI Upper	CI Width
North East	5.300%	155	3.010%	10.533%	7.522pp
North West	6.400%	422	4.436%	9.151%	4.715pp
Yorkshire & the Humber	4.700%	273	2.609%	7.654%	5.045pp
East Midlands	4.500%	318	2.563%	7.128%	4.565pp
West Midlands	4.500%	343	2.688%	7.122%	4.434pp
East of England	5.800%	348	3.555%	8.417%	4.862pp
London	5.900%	332	3.518%	8.500%	4.982pp
South East	4.100%	549	2.897%	6.338%	3.441pp
South West	3.500%	318	2.118%	6.388%	4.270pp

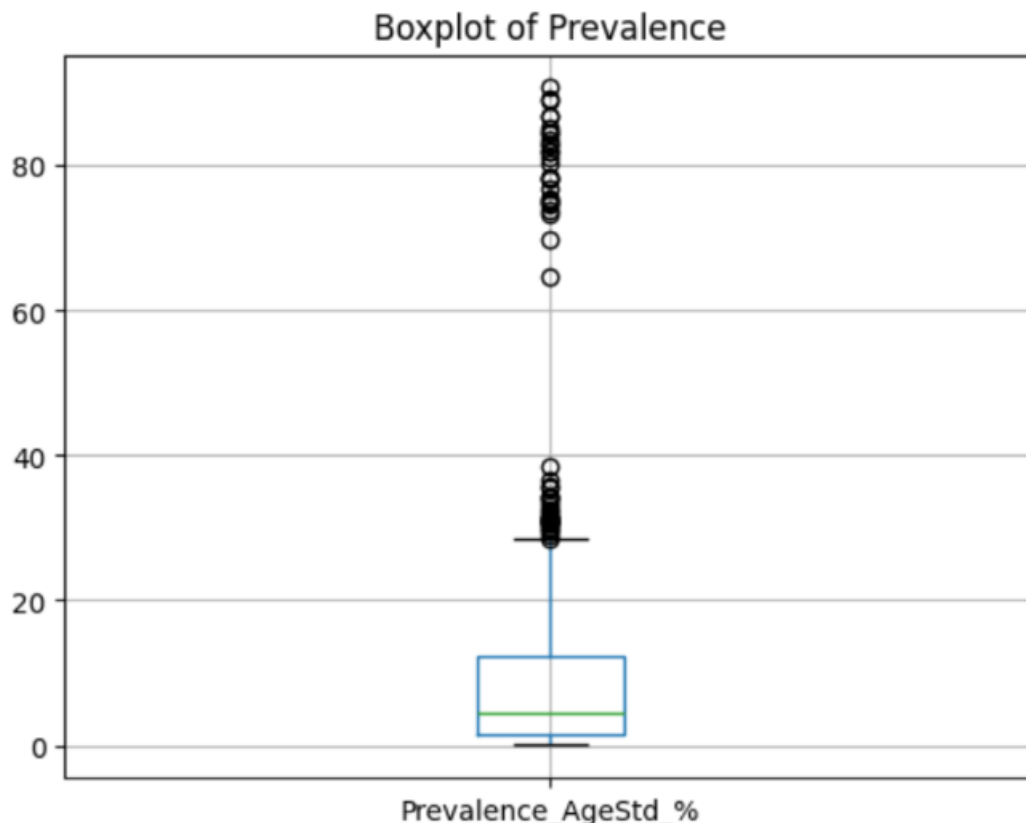
PTSD					
Region	Prevalence%	n	95% CI Lower	95% CI Upper	CI Width
North East	33.121%	134	25.441%	41.147%	15.706pp
North West	31.565%	387	27.406%	36.650%	9.245pp
Yorkshire & the Humber	24.331%	239	18.265%	28.911%	10.646pp
East Midlands	38.397% ▲	293	32.843%	43.903%	11.060
West Midlands	32.664%	301	28.046%	38.622%	10.575pp
East of England	30.545%	335	24.515%	34.202%	9.687pp
London	28.395%	291	23.375%	33.667%	10.292pp
South East	34.532%	509	31.357%	39.637%	8.280pp
South West	31.233%	292	27.130%	37.791%	10.661pp

PSYCHOTIC DISORDER					
Region	Prevalence%	n	95% CI Lower	95% CI Upper	CI Width
North East	nan%	327	nan%	nan%	nanpp
North West	0.339%	886	0.116%	0.992%	0.876pp
Yorkshire & the Humber	0.297%	588	0.075%	1.163%	1.088pp
East Midlands	0.642%	634	0.252%	1.627%	1.375pp
West Midlands	0.281%	674	0.075%	1.051%	0.976pp
East of England	1.075%	709	0.537%	2.141%	1.603pp
London	1.077% ▲	641	0.520%	2.217%	1.697
South East	0.128% ▼	1027	0.027%	0.600%	0.573
South West	nan%	638	nan%	nan%	nanpp

BIPOLAR DISORDER					
Region	Prevalence%	n	95% CI Lower	95% CI Upper	CI Width
North East	1.251%	138	0.282%	4.697%	4.415pp
North West	2.188%	402	1.132%	4.106%	2.974pp
Yorkshire & the Humber	0.808%	245	0.222%	2.917%	2.696pp
East Midlands	2.026%	309	0.885%	4.154%	3.269pp
West Midlands	2.390%	316	1.031%	4.408%	3.377pp
East of England	3.298%	340	1.919%	5.877%	3.958pp
London	2.713%	303	1.602%	5.606%	4.004pp
South East	1.785%	526	0.867%	3.152%	2.285pp
South West	2.140%	302	0.901%	4.240%	3.339pp



7. **Outlier Detection** : The box plot graph I gave for Distribution of Prevalence Rates explains how the mental health prevalence values are spread across the dataset.



8. **Sex-wise Comparison** : Prevalence differences may exist between male and female populations.

```
Sex
All adults    10.163908
Men           10.195189
Women         10.500399
Name: Prevalence_AgeStd_%, dtype: float64
```

9. **Correlation Analysis** : Check relationships between numerical variables.

```
          Prevalence_AgeStd_%  Sample_size  Standard_error% \
Prevalence_AgeStd_%          1.000000    -0.040884         0.593847
Sample_size                -0.040884         1.000000        -0.451560
Standard_error%             0.593847        -0.451560         1.000000
CI_Width_pp                 0.593849        -0.451560         1.000000

          CI_Width_pp
Prevalence_AgeStd_%    0.593849
Sample_size            -0.451560
Standard_error%         1.000000
CI_Width_pp            1.000000
```

- larger sample size → smaller standard error

Correlation analysis revealed a moderate positive relationship between prevalence estimates and both standard error and confidence interval width. Sample size showed a moderate negative correlation with uncertainty measures, indicating that larger samples produced more precise prevalence estimates. A perfect positive correlation was observed between standard error and confidence interval width due to their direct statistical relationship.

10. Composite Mental Health Burden Score: No single disorder gives us full idea.

To rank regions by overall mental health burden, So:

- Take the observed prevalence for each disorder in each region (All Adults only).
- Standardise each disorder to a z-score: $z = (\text{region_value} - \text{national_mean}) / \text{national_std}$.

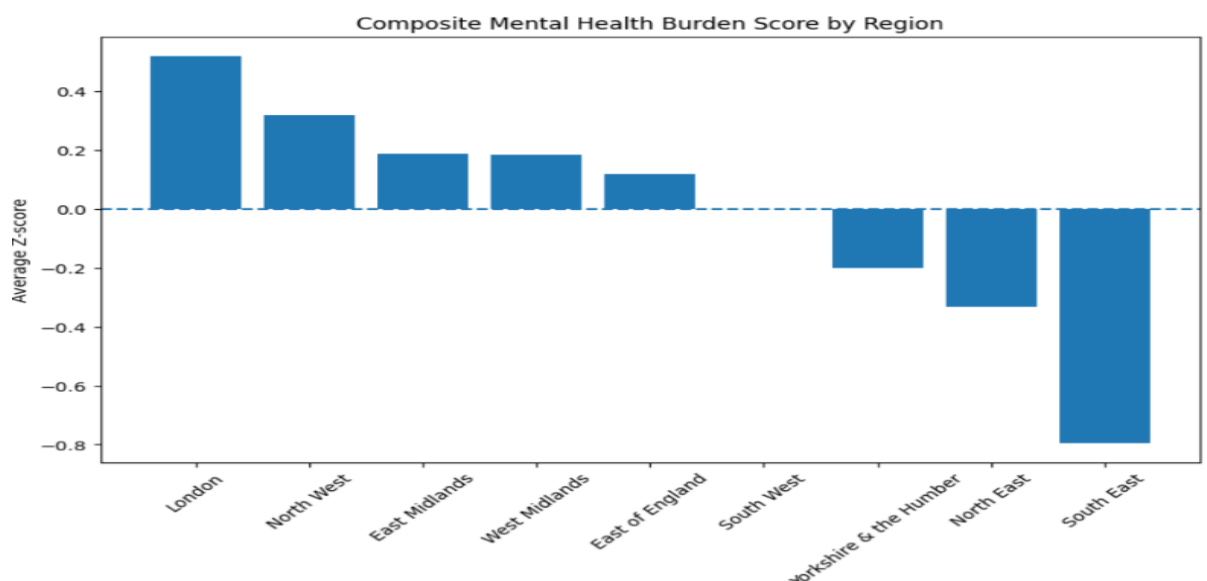
- Average the z-scores across all disorders for each region.
- A positive score means above-average burden; negative means below average.

	Region	Composite_Burden_Score	Population \
2	London	0.519	6806412.0
4	North West	0.320	5790512.0
0	East Midlands	0.186	3782353.0
7	West Midlands	0.183	4599064.0
1	East of England	0.119	4875281.0
6	South West	0.001	4473319.0
8	Yorkshire & the Humber	-0.202	4347165.0
3	North East	-0.332	2154854.0
5	South East	-0.794	7184102.0

Interpretation	
2	Above Average Burden
4	Above Average Burden
0	Above Average Burden
7	Above Average Burden
1	Above Average Burden
6	Above Average Burden
8	Below Average Burden
3	Below Average Burden
5	Below Average Burden

Interpretations:

- 0 → England average.
- Positive → above average burden.
- Negative → below average.



11. Conclusion: This study developed a Statistical Baseline Model (SBM) to examine the prevalence of mental health disorders across regions using prevalence estimates, confidence intervals, standard errors, and regional burden analysis. Exploratory statistical analysis showed that mental health prevalence varies considerably across both disorders and geographic areas, indicating unequal distribution of mental health burden within the population.

Confidence interval analysis using prevalence proportions demonstrated that regions with larger sample sizes generally produced narrower confidence intervals and lower standard errors, indicating more reliable prevalence estimates. Correlation analysis further confirmed that sample size had a negative relationship with uncertainty measures such as standard error and confidence interval width, while prevalence showed a moderate positive relationship with statistical uncertainty.

- **South East** and **North East** show comparatively lower burden levels.
- CMD (anxiety, depression) is the most dominant contributor to overall mental health issues.
- **Suicidal thoughts** are consistently **high** across all regions, indicating a national concern.
- Substance use disorders (alcohol, drugs) are more prominent in urban regions.
- Low-prevalence disorders (**psychotic, bipolar**) are less common but clinically critical.
- Confidence intervals confirm that results for common disorders are statistically reliable.
- Composite **Z-score** analysis validates consistent patterns of regional mental health burden.
- Dataset or Report : [Statistical Model Report \(APMS 2014 MH Regional\)](#)